

Task Groups	Task Clusters	Task Definition	Task Input	Task Output
Graph Structure Modeling	Connection Detection, Cycle Detection, Hamilton Path, Bipartite Matching, Shortest Path, Degree Computing	The tasks in this group aim to make LLMs better understand some basic graph structures. The input only contains nodes, directed or un-directed edges, and optional weights.	$\mathcal{X}_i = [\mathcal{I}_i, \mathcal{C}_i]$	$\mathcal{Y}_i = \mathcal{A}_i$
Graph Language Modeling	Graph Caption Generation	The task aims to generate a caption passage $\mathcal{P}_i$ to describe the graph $\mathcal{G}_i$.	$\mathcal{X}_i = [\mathcal{I}_i, \mathcal{C}_i]$	$\mathcal{Y}_i = \mathcal{P}_i$
	Graph Question Answering	The task aims to reason on the whole graph $\mathcal{G}_i$ and find an entity as the final answer $\mathcal{A}_i \in \mathcal{E}_i$.	$\mathcal{X}_i = [\mathcal{I}_i, \mathcal{C}_i, \mathcal{P}_i]$	$\mathcal{Y}_i = \mathcal{A}_i$
	Graph Node Classification	The task aims to classify the target node into pre-defined classes based on $\mathcal{G}_i$.	$\mathcal{X}_i = [\mathcal{I}_i, \mathcal{C}_i, \mathcal{P}_i]$	$\mathcal{Y}_i = \mathcal{A}_i$
	Graph Link Prediction	The task aims to predict the relation between two given nodes based on $\mathcal{G}_i$.	$\mathcal{X}_i = [\mathcal{I}_i, \mathcal{C}_i, \mathcal{P}_i]$	$\mathcal{Y}_i = \mathcal{A}_i$
	Graph Relevance Inspection	The task aims to detect whether the graph $\mathcal{G}_i$ is relevant to the passage $\mathcal{P}_i$, we have $\mathcal{A}_i \in \{\text{relevant, irrelevant}\}$.	$\mathcal{X}_i = [\mathcal{I}_i, \mathcal{C}_i, \mathcal{P}_i]$	$\mathcal{Y}_i = \mathcal{A}_i$
	Graph Collaboration Filtering	The task aims to predict whether the target user prefers the target item based on the whole graph $\mathcal{G}_i$, the answer $\mathcal{A}_i$ can be set as a score.	$\mathcal{X}_i = [\mathcal{I}_i, \mathcal{C}_i, \mathcal{P}_i]$	$\mathcal{Y}_i = \mathcal{A}_i$
Graph Generation Modeling	Knowledge Graph Generation	The task aims to given a passage $\mathcal{P}_i$ that describes a piece of factual or commonsense information, the task aims to extract entities and relations from $\mathcal{P}_i$ to generate a graph $\mathcal{G}_i$.	$\mathcal{X}_i = [\mathcal{I}_i, \mathcal{P}_i]$	$\mathcal{Y}_i = \mathcal{C}_i$
	Structure Graph Generation	The task aims to generate a graph to meet the structure information described in the passage $\mathcal{P}_i$.	$\mathcal{X}_i = [\mathcal{I}_i, \mathcal{P}_i]$	$\mathcal{Y}_i = \mathcal{C}_i$
Graph Thought Modeling	Arithmetic Symbolic Robotic Logic	The task aims to solve the general reasoning task in three think steps: 1) first find the question subject, 2) then generate a thought graph $\mathcal{G}_i$ to express the rationale and 3) finally output the result $\mathcal{A}_i$ based on the graph.	$\mathcal{X}_i = \mathcal{I}_i$	$\mathcal{Y}_i = [\mathcal{C}_i; \mathcal{A}_i]$